

EXP 01. Find S Algorithm:

```
import pandas as pd
import numpy as np

# Student Placement Dataset
data = {
    'CGPA': ['High', 'High', 'Medium', 'High', 'Medium', 'High'],
    'Communication': ['Good', 'Average', 'Good', 'Good', 'Average', 'Excellent'],
    'Internship': ['Yes', 'Yes', 'No', 'No', 'Yes', 'Yes'],
    'Placement': ['Yes', 'Yes', 'No', 'Yes', 'No', 'Yes']
}

df = pd.DataFrame(data)

print("Register Number : 192321113")
print("\nTraining Data:")
print(df)

concepts = np.array(df.iloc[:, :-1])
target = np.array(df.iloc[:, -1])

# Initialize hypothesis
hypothesis = concepts[0].copy()

# Find-S Algorithm
for i in range(len(target)):
    if target[i] == "Yes":
        for j in range(len(hypothesis)):
            if hypothesis[j] != concepts[i][j]:
                hypothesis[j] = "?"

print("\nFinal Hypothesis:")
print(hypothesis)
```


Final Hypothesis:
['High' '?' '?']